

Teaching Statement

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My goal as a statistics instructor is not simply to teach students how to fit a model, but to help them understand *why* a method is needed, *when* it is valid, *how* it is implemented, and *what* its output means in the context of a scientific or practical question. I therefore organize my teaching around a progression from **motivation and statistical reasoning to computation, interpretation, and application**. This philosophy has developed through teaching students at very different levels ranging from introductory probability and statistics to graduate categorical data analysis, time series, and causal inference, and through learning from student feedback as I have moved from teaching assistant to independent course designer, instructor, and research mentor.

Teaching through motivation, reasoning, computation, and interpretation

Statistics becomes meaningful when students can see the question before they see the formula. I typically begin a new topic with a concrete problem: What does an odds ratio actually tell us? Why can two forecasting models with similar in-sample fit behave differently out of sample? Why does regression adjustment not automatically make an observational comparison causal? Once the question is clear, I develop the statistical reasoning and assumptions behind the method, connect the theory to computation in R or Python, and return to interpretation. My aim is for students to leave a course able not only to execute an analysis but also to explain and defend the choices that produced it.

This approach is especially important in graduate courses, where rigor and accessibility have to coexist. I want students to see enough mathematical structure to understand what a method is doing without allowing derivation to become an end in itself. In Applied Categorical Data Analysis, for e.g., I connect likelihood-based models, coefficient interpretation, diagnostics, and real-data analysis rather than treating them as separate topics. One student described this as my “ability to explain complex statistical concepts clearly while keeping the focus on interpretation rather than just technique.” Another wrote that the lectures “consistently grounded the theory in interpretation and application.” These comments capture the balance I try to achieve: technical depth should sharpen applied understanding rather than obscure it.

Computation is integrated into that intellectual structure rather than treated as a separate laboratory skill. Students use code to simulate data, examine assumptions, compare estimators, visualize uncertainty, and explore what happens when ideal conditions fail. In Time Series Forecasting, code accompanied the statistical ideas throughout the course. Students described the “example codes and lectures” as particularly helpful and appreciated that lectures, code, and recordings were made readily available. One student noted that the review quizzes clarified which concepts would be important in later assignments, while another emphasized that I “made the concepts applicable” and explained “what is going on” rather than simply presenting procedures. My goal is for students to become confident enough to adapt an analysis to a new problem rather than simply reproduce a template.

Curriculum development and reflective course design

A defining feature of my first three semesters as a faculty member has been curriculum development. During that period, I developed **two new graduate courses** and **completely redesigned a third**. I developed **Time Series Forecasting**, combining stochastic-process foundations with modern forecasting practice, and **Causal Inference**, organized around potential outcomes, causal graphs, identification, confounding, instrumental variables, difference-in-differences, mediation, transportability, and computational applications. I also substantially redesigned **Applied Categorical Data Analysis**, rebuilding the lecture materials, examples, assessments, and computational components around a coherent progression from binary and multinomial models to correlated categorical outcomes.

Developing these courses has reinforced my view that course design is itself an iterative, evidence-based process. My early graduate courses were ambitious in breadth and technical depth. Students were positive about my preparation, subject knowledge, organization, accessibility, and willingness to answer questions, but some also told me that the pace could be fast and that technically dense topics needed stronger signposting, more guided coding, and additional real-world examples. I responded by organizing lectures around fewer central takeaways, adding recaps, increasing guided computational examples, and distinguishing more explicitly between foundational ideas and extensions. Students noticed these changes: one wrote that “the recaps at the beginning of each lecture and the homework solutions were very helpful,” while another commented that I “listen to our suggestions and make immediate changes to improve the lectures.” I regard this cycle: teach, gather evidence, reflect, revise, as an essential part of effective instruction.

I also place considerable value on accessibility outside the classroom. I encourage questions, maintain regular office hours, respond promptly to student emails, and make detailed notes, code, and supporting materials available so that students can revisit difficult ideas. Students have repeatedly described me as approachable, responsive, and supportive; one Time Series student wrote that I “made complex concepts accessible and engaging” and was “approachable, responsive to questions, and clearly passionate about the subject.” This matters particularly in statistics, where students often enter the same classroom with very different mathematical, computational, and disciplinary preparation.

Mentoring students toward independent scholarship

I view research mentoring as an extension of teaching. As Rabindranath Tagore wrote, “A teacher can never truly teach unless he is still learning himself.” That idea resonates strongly with my approach to mentoring: the most productive student projects are collaborative learning experiences in which I provide structure and technical guidance while gradually transferring ownership of the scientific question to the student.

At Kennesaw State, I have mentored students at the undergraduate, master’s, and doctoral levels and have tried to build a genuine research pipeline rather than a collection of isolated projects. Through the **First-Year Scholars Program**, I mentored three undergraduate researchers. Two of their abstracts were accepted for presentation at the National Conference on Undergraduate Research (NCUR), and work from the group produced an ICMLA short paper, a manuscript that received a favorable revision from the *American Journal of Undergraduate Research*, and another manuscript that is ready for submission. Two of these students have continued with me through the **Sophomore Scholars Program**. I see that continuity as especially important: undergraduate research becomes transformative when students move beyond completing assigned tasks and begin to recognize themselves as contributors to scholarship.

My mentoring at the graduate level follows the same progression toward independence. I supervised two master’s students: one project led to a peer-reviewed paper published in the *DATA 2026* conference proceedings, while a second manuscript is nearing submission. One of these students subsequently entered the Ph.D. program and continues to work with me, and I welcomed an additional Ph.D. student this semester. In a doctoral internship evaluation, one mentee wrote that I am “very rich in mathematical derivation and simulation, which has shaped my research.” I value that comment because it reflects what I hope mentoring accomplishes: not merely completing a paper, but helping a student acquire ways of thinking and working that shape future research.

Early in a project, I typically provide a structured reading plan, reproducible code, simulation exercises, and short-term milestones. As students gain confidence, I shift responsibility toward study design, methodological choices, interpretation, writing, and presentation. My goal is for students to leave the mentoring relationship able to formulate a question, identify the appropriate tools, recognize limitations, communicate results clearly, and eventually mentor others themselves.

Area	Summary
Courses taught	Introduction to Probability and Statistics; Applied Categorical Data Analysis; Time Series Forecasting; Causal Inference; with earlier teaching experience in Probability and Statistics for Engineers, Statistical Theory, and Machine Learning.
Course development	Developed two new graduate courses: Time Series Forecasting and Causal Inference within my first three semesters as faculty, and completely redesigned Applied Categorical Data Analysis , including lectures, computational examples, assessments, and course organization.
Courses comfortable teaching	Introductory and mathematical statistics; probability; regression and generalized linear models; categorical data analysis; time series; causal inference; longitudinal data analysis; survival analysis; biostatistics; computational statistics; statistical machine learning; and data science.
Mentoring achievements	Mentored undergraduate, master’s, and doctoral researchers. Student work has led to NCUR presentations, an ICMLA short paper, a DATA 2026 proceedings paper, a favorable AJUR revision, and additional manuscripts nearing submission. Two First-Year Scholars continued into the Sophomore Scholars Program, and one master’s student continued into the Ph.D. program.
Future teaching and mentoring goals	Continue building a vertically integrated undergraduate to Ph.D. research pipeline; expand project-based and reproducible-computing components across statistics courses; and develop advanced offerings in longitudinal and survival methods, modern biostatistics, and statistical methods for biomedical and high-dimensional data.